SU(2) Gauge-Coupled Higgs Hessian

Gauge–Goldstone Mixing and Ghost-Sector Diagnostic

In Vero Vacuo — M. S. Cook — September 30, 2026

The scalar Hessian is extended in background 't Hooft–Feynman gauge ($\xi = 1$). For each of the three SU(2) generators, the matrix-free solver acts on $(A_1, A_2, A_3, A_4, \chi)$ with

$$\mathcal{H}_{A\chi} = \begin{pmatrix} -\Delta + m_W^2 & 2q\,\partial_\mu\rho \\ 2q\,\partial_\nu\rho & -\Delta + V_T'' + m_W^2 \end{pmatrix}, \qquad q = \frac{g}{2}, \qquad m_W^2 = q^2\rho^2.$$

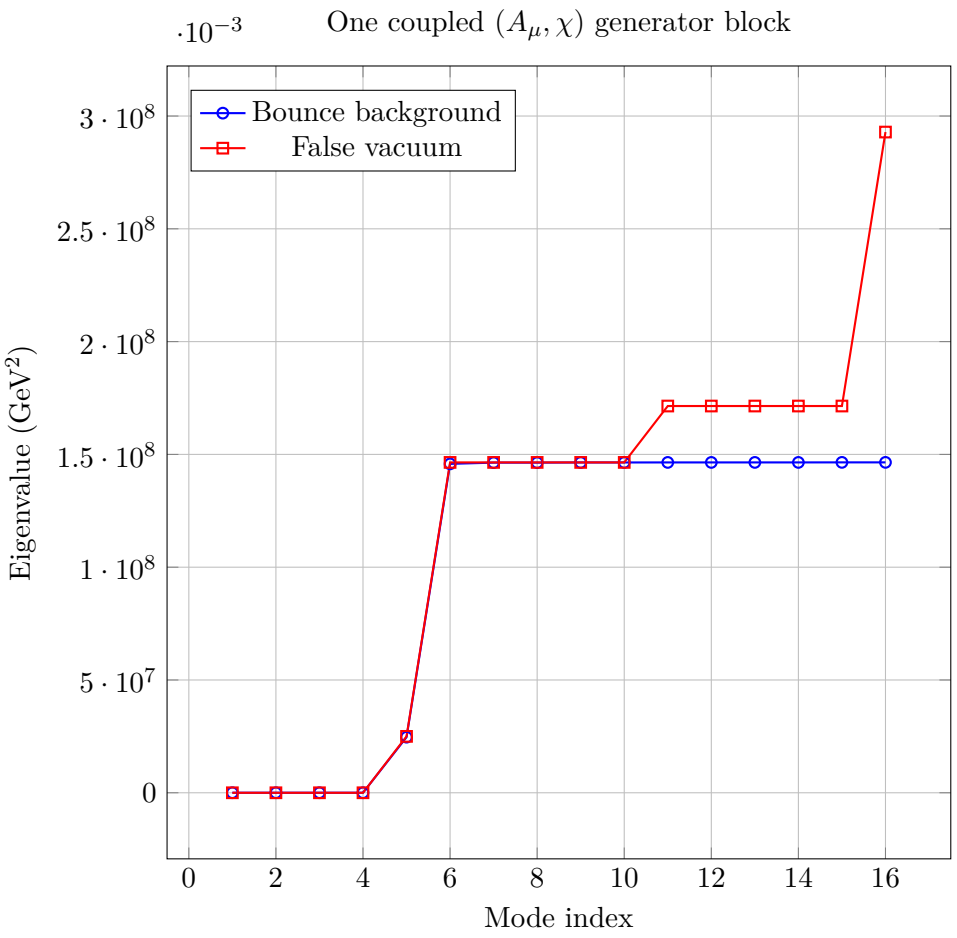
Together with the radial Higgs block, the three gauge–Goldstone blocks form a $16L^4$ bosonic Hessian. The Faddeev–Popov ghost operator is evaluated separately because it contributes with a different determinant power.

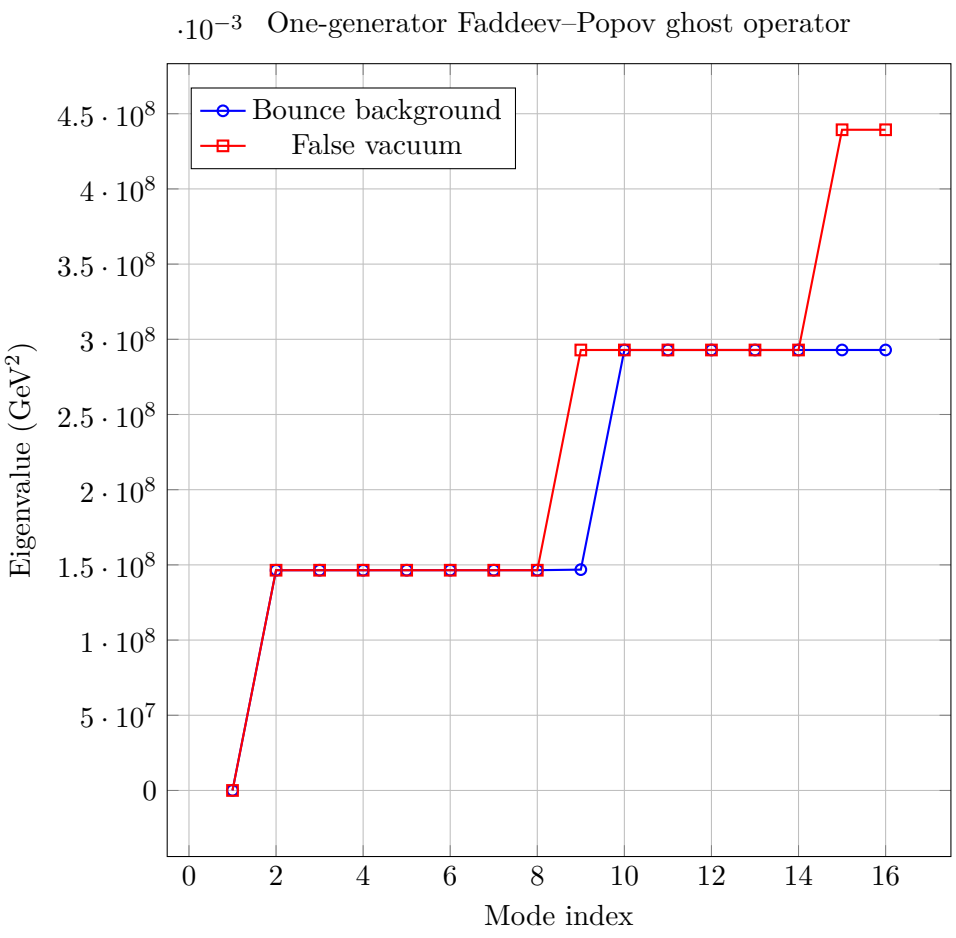
Quantity	Setting	Result
Lattice	$L = 8$, $a = 2.0 \times 10^{-3}\,\mathrm{GeV}^{-1}$	4096 sites
Background	Contained Gaussian diagnostic	boundary fraction 1.495×10^{-5}
Gauge coupling	$g = 0.65$, $\xi = 1$	$q = g/2$
Bosonic Hessian	Higgs plus three (A_μ, χ) blocks	65536×65536
Gauge spectral window	One representative generator	16 modes
Generator multiplicity	Radial background	3 identical blocks
Self-adjointness check	Bilinear-form residual	2.26×10^{-16}

Scope. This calculation validates the coupled lattice Hessian and its low spectrum. The Gaussian background is a contained diagnostic profile, not a stationary Coleman bounce. The result is therefore not yet a physical gauged decay prefactor. Gauge/ghost determinant combination, collective-mode treatment, renormalization, and continuum checks remain required.

Source: `Higgs_fluctuation_solver_stabilized.py`.

SU(2) Gauge–Goldstone and Ghost Spectra





The three SU(2) generator blocks are identical for the radial background. The four near-zero false-vacuum gauge levels are the constant periodic gauge configurations. They require a gauge-volume prescription before any determinant ratio can be interpreted physically.